

NH3Bench: LLM Agents as Real-Time Operators of an Industrial Ammonia Refrigeration Plant

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Abstract

We demonstrate NH3Bench, an executable benchmark for a general question: can a language-model agent hold real-time supervisory control of an industrial installation in complex risk-related scenarios? Three properties define the task: the process evolves whether or not the agent acts (inaction ends in catastrophe); it does not pause awaiting decisions (reasoning tokens convert to seconds, so a correct answer can arrive later than necessary); and it is only partially digitalized (so the agent should give orders to workers for some actions).

NH3Bench reproduces this situation using a physics-based digital twin of an industrial ammonia (R717) refrigeration plant and consider the agent is the shift engineer. Across six calibrated scenarios, nine LLMs score between 18.4 and 87.9 of 100 that highlights the limitations of existing agents for industrial tasks. During the demo provided in <>, each scenario could be examined and reproduced in interactive way, as well as solved by expert manually.

Introduction

Language-model agents are being evaluated for operational roles in industrial facilities. However, existing evaluations surround this task without closing its loop. Digital-native agent benchmarks score tool use, dialogue under rule compliance, and procedure following (Liu et al. 2023; Yao et al. 2024; Nandi et al. 2025); industrial suites diagnose over logged sensor data—no actuation, no physics integrating in time (Patel et al. 2025); LLM-plus-twin control frameworks close a physical loop but as demonstrations, not benchmarks (Vyas and Mercangöz 2024); process-control gyms target RL (Bloor et al. 2024); operator studies run text simulators without physics (Lee et al. 2026). In (Zheng et al. 2025), fuse time burned per token (Parekh et al. 2026), but not as a safety variable where correct-but-late means a failure. The proposed benchmark differs in structural properties. The dynamics are autonomous: in a developing incident *doing nothing is itself a decision* with physical consequences, so there is no safe default. Deliberation is not free: the plant integrates forward while the agent thinks, so correctness and latency cannot be scored separately. And no SCADA sees everything: the information set includes sight glasses, local

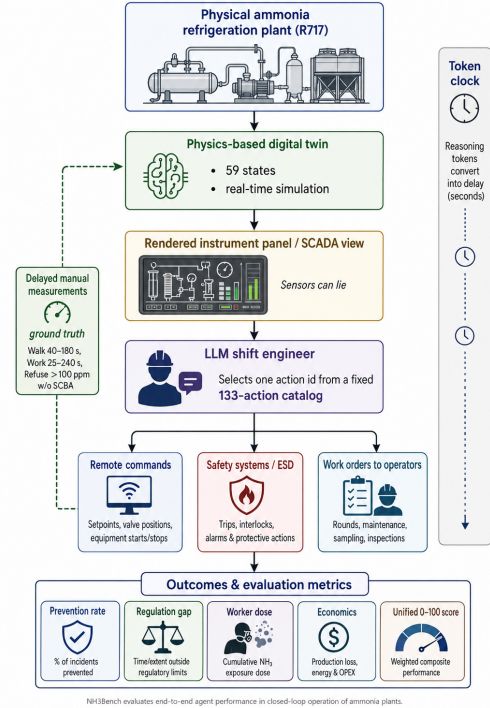


Figure 1: The NH3Bench design.

gauges, smells and sounds, extended only by dispatching people at a cost in minutes and, sometimes, exposure. No prior benchmark operationalizes all three at once; NH3Bench’s claim is the combination, not any single mechanism.

NH3Bench implements this idea using case study is industrial ammonia (R717) refrigeration, chosen for its documented accident record (e.g. the 2010 Millard Refrigerated Services ammonia release was caused by a defrost cycle interrupted by a power loss (U.S. Chemical Safety and Hazard Investigation Board 2015), used as an inspiration for Scenario 1). Every outcome is produced by a thermodynamic model of system.

policy	S1	S2	S3	S4	S5	S6	mean
oracle (scripted)	100	100	100	100	100	100	100.0
Fable 5	100	100	80	100	78	69	87.9
Astra (med.)	100	100	80	100	40	69	81.5
Opus 5	100	100	80	100	67	0	74.6
Sol (med.)	100	0	80	100	62	69	68.5
GLM-5.3 (max.)	0	89	75	68	83	69	64.1
always-ESD	73	95	57	95	57	0	62.8
Sonnet 5	0	0	80	53	84	69	47.6
Haiku 4.5	0	95	80	0	94	0	44.9
written regulation	0	95	57	0	94	0	41.0
Terra (med.)	0	0	75	100	50	0	37.5
inaction	0	0	80	0	100	0	30.1
Luna (med.)	0	0	80	0	30	0	18.4

Table 1: Unified score, twin seed 1. Complete per-decision transcripts are published in the repository.

Benchmark Design

The interface is domain-independent: rendered instrument readings in, one action id from a fixed catalog out, one clock shared with the plant (Figure 1). The ammonia case study fills this interface with physics: the twin models a 250 t/day dairy plant carrying a 4,170 kg R717 charge—compressors on two suction stages, evaporative condensers, hot-gas defrost, an ice-water circuit, HACCP product temperatures, and ammonia dispersion with per-operator toxic-dose accounting.

Six scenarios are admitted by a calibration quartet plus an always-ESD baseline: inaction must end in catastrophe; a scripted oracle must pass cleanly on observable evidence; the checklist policy (π_{reg}) must fail or pay heavily; random action must not save. Mechanisms include a hung defrost silently swallowing its remote abort (S1), a loud leak masking a dead level sensor (S2), an air-in-condenser false trail (S3), and an isolation trap where the regulation’s own reflex causes the rupture (S4). Two scenarios are *mirrored controls*: S5 punishes over-reaction to a drifting gas detector, while S6 presents the same evidence and then lets the discredited instrument start telling the truth—the fatal error is closing the question for good. S6 also closes a loophole: “emergency-stop everywhere” scores 88.6/100 on S1–S5 with no diagnosis, but S6’s leak is independent of the compressors and ends in catastrophe.

The metrics, too, attach to the task, not the domain: alongside Prevention Rate the benchmark reports *Regulation Gap*—score minus that of the published checklist policy; an agent that cannot beat the written regulation scores below zero. A unified 0–100 score multiplies independent axes: catastrophe or an incapacitated worker zeroes the run; operator dose multiplies (never monetized); economics and discipline fill the remainder. Points of no return, found by bisection over delayed-oracle starts, separate “did not understand” from “understood too slowly.”

Measured Results

Table 1 summarizes nine language models against the reference policies. Rankings invert on the mirrored S5/S6 pair, so no fixed disposition survives both. The token clock also

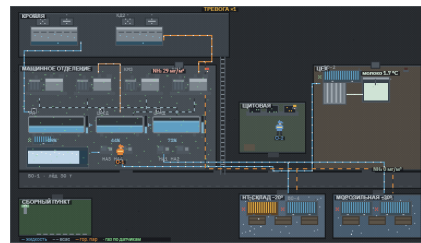


Figure 2: The trainer’s plant view 33 minutes into S2 of the ammonia case study: the machine-room NH_3 detector reads 29 mg/m^3 , the alarm banner flashes, the drum level sensor reads an impossible 73 %, and operator O-1 walks to the local level glass.

makes verbosity operational: a correct action arriving after a point of no return is still a failure.

One GLM-5.3 run at maximum reasoning scored 64.1 (PR 0.833, Regulation Gap +23.1); failure on S1 left its worst-scenario score at zero, so this remains a single-run observation.

Three Terra sensitivity checks qualify the single-run ranking. At fixed medium reasoning, ten complete runs over twin seeds 1–10 averaged 42.1 (SD 8.2, range 35.1–55.0; bootstrap 95% CI [37.5, 47.3]); most variation came from rarely preventing S1 or S2. At twin seed 1, five independent samples averaged 36.92 (SD 0.58, range 36.5–37.6; bootstrap 95% CI [36.50, 37.36]), showing much smaller observed model-sampling dispersion. Across all six reasoning levels, three runs per level produced means from 37.2 to 51.0, but levels were not distinguishable (two-sided 100,000-permutation Kruskal–Wallis $H = 2.511$, $df = 5$, $p = 0.817$), with no ordered trend (Spearman $\rho = -0.116$, permutation $p = 0.649$). Because model-sampling RNG was uncontrollable, the multi-seed study measures end-to-end variability, whereas fixed-twin repeats remove simulator perturbations; the small reasoning study does not establish equivalence.

Demonstration

The demonstration has three phases:

1. **Watch:** follow the panel, clock, work orders, and model reasoning during a chosen live episode.
2. **Play:** operate the same Pyodide twin in the expert-validated browser trainer (Figure 2); the score joins a leaderboard seeded with Table 1.
3. **Inspect:** examine decision transcripts aligned to the trajectory.

Conclusion

NH3Bench makes real-time supervisory control by a language agent measurable. Its combination of irreversible consequences, priced deliberation, and information bought with human exposure is not specific to ammonia. The calibrated-twin recipe transfers to other installations with faithful twins; code, policies, metrics, and transcripts are public in the repository.

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